

Requirement Analysis Phase

Solution Requirements

Date	18th June 2026
Team ID	
Project Name	MediCareBook (Book a Doctor)
Maximum Marks	4 Marks

Section 1: Functional Requirements (FR)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Identity & Access Control	* Secure sign-up/logins with email format checks. * Cryptographic password encryption utilizing BcryptJS salting. * Custom Role-Based Access Control middleware to prevent routing bypass.
FR-2	Doctor Directory Search	* Directory lookup displaying only approved, vetted practitioner cards. * Multi-option specializations filter selector dropdown. * Case-insensitive doctor name lookup input bar.
FR-3	Patient Booking Engine	* Form interface enabling date and slot times selection. * Saving appointment bookings to database collection with default status set to pending. * Consolidated personal ledger showing booking status changes.
FR-4	Practitioner Dashboards	* Indicator panels showing scheduling counts (Pending, Approved, Completed). * Dynamic booking action queues with Approve and Decline control buttons. * Form editor to modify consulting fees and hour grids.
FR-5	Admin Vetting & CRUD	* Administrative verification queue to audit credentials and toggle doctor status to approved. * Accounts lookup panel with cascade deletion triggers.

Section 2: Non-Functional Requirements (NFR)

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Fully responsive layout styled with Tailwind CSS utility classes, optimized for desktop, tablet, and mobile screens. Under 3-click booking paths.

NFR No.	Non-Functional Requirement	Description
NFR-2	Security	Password hashing using BcryptJS (salt factor of 10). Route evaluation middleware blocks horizontal access escalations.
NFR-3	Reliability	Mongoose model validations ensure correct field inputs, preventing malformed document insertion. Global client Axios error handlers catch failed requests.
NFR-4	Performance	Vite packaging compiles small static assets loading in under 1.5 seconds. Database index keys on email addresses keep query processing under 200ms.
NFR-5	Availability	Stateless backend server instances are suitable for cloud clustering to reach a target 99.9% application uptime.
NFR-6	Scalability	Decoupled frontend/backend, allowing isolated scaling of Node nodes. MongoDB scales horizontally through replica sets.